

Speeding Up Data Visualisation with Julia

Today's real-time applications deal with large volumes of data. The mammoth size of the data makes it mandatory to have powerful visualisation tools. This article provides an introduction to visualisation in Julia. As the syntax in Julia is very user friendly, the visualisation process can be understood fairly easily.



Software applications consume data. The user benefits only if this data is rendered in a recognisable format. As the volume of data being fed to applications is rising exponentially, rendering the data in a human-understandable format becomes an important aspect of building any successful application software. Hence, data visualisation has become a powerful tool for app builders. There are many libraries to ease this complex task of rendering the data in an easy-to-infer, graphical representation. Almost all of the modern day programming languages have libraries to perform this task.

Julia is a modern programming language that successfully combines the speed of C with the ease of Python. It is becoming popular in developer circles as it has the capabilities to perform various tasks without burdening programmers. Let's look at some of the prominent visualisation libraries of Julia (Figure 1).

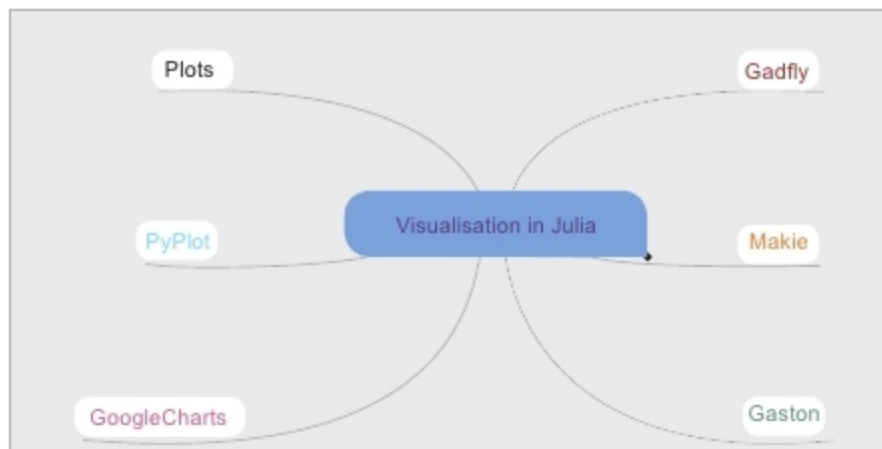


Figure 1: Visualisation libraries in Julia

Gadfly

Gadfly is an important data visualisation library in Julia. The Gadfly visualisation system is coded in the Julia programming language. The official document states that the Gadfly is inspired by the book, 'The Grammar of Graphics'.

The prominent features of Gadfly are listed below:

- The graphics rendered through Gadfly are of publication quality. It supports PNG, SVG, PDF and Postscript.
- Gadfly works without any additional configuration requirements in Julia.
- Another major advantage of Gadfly is that it has been tightly integrated with *DataFrames.jl*.
- Dynamic visualisation abilities such as zooming and panning are important features of Gadfly.

A gallery of visualisation that's been provided in the documentation is shown in Figure 2.

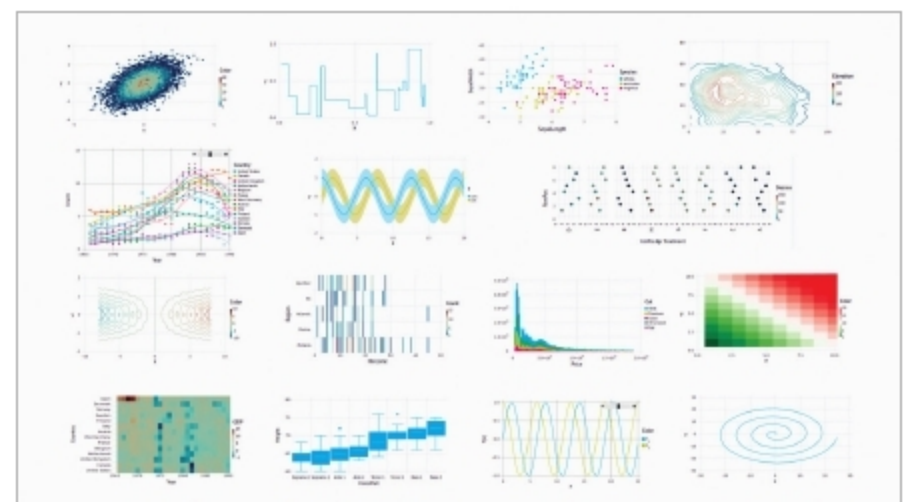


Figure 2: Julia visualisation gallery